DO NOW — RECALL

OOP debugging with IDEs

Name: _____ Class: _____ Date: _____

1. Which structure supports BFS traversal?

2. What is an object's field?

3. When a local variable shadows a field, which name is used without qualification?

4. Why might Step Into be more useful than Step Over for a faulty function?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A FIFO queue.

2. Model answer

Stored state belonging to that object.

3. Model answer

The local variable.

4. Model answer

It lets you inspect statements, parameters and field updates inside the called function.

Vocabulary — know these terms

breakpoint — A chosen instruction where execution pauses.

Step Into — Enter a called function for debugging.

Step Over — Execute a call without entering it line by line.

watch — IDE display of chosen variables/expressions.

object state — Current values of an object's fields and properties.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model the reasoning

The method always assigns true, so the second call fails. Fix to IsOn = !IsOn; and test twice.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Set a breakpoint

Pause at a method call or the following instruction to inspect current state.

Step into to inspect internals

Step Into executes the called method one instruction at a time. Watch Count and parameter n.

Step over to skip internals

Step Over runs the entire call and pauses at the next statement, allowing you to inspect the result.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

BEFORE Add(4)

STATE / VARIABLE	VALUE
Count	0
n	—

Constructor leaves Count=0.

BEFORE Add(4) Count equals 0; n equals —. Constructor leaves Count=0.

New Learning

STEP INTO Add

STATE / VARIABLE	VALUE
Count	0
n	4

Observe local parameter n=4.

STEP INTO Add Count equals 0; n equals 4. Observe local parameter n=4.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Lamp starts IsOn=false. Toggle() sets IsOn=true on every call. Diagnose the second-call error.

After the first call, true is correct; after the second, IsOn should become false.

FIX

STATE / VARIABLE	VALUE
IsOn	false
code	IsOn = !IsOn

Toggling twice restores original state.

FIX IsOn equals false; code equals IsOn = !IsOn. Toggling twice restores original state.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Lamp starts IsOn=false. Toggle() sets IsOn=true on every call. Diagnose the second-call error.

After the first call, true is correct; after the second, IsOn should become false.

CALL TWICE

STATE / VARIABLE	VALUE
IsOn	true
expected	false

State fails to toggle back.

CALL TWICE IsOn equals true; expected equals false. State fails to toggle back.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

AFTER Add

STATE / VARIABLE	VALUE
Count	4
n	out of scope

Watch window shows Count=4.

AFTER Add Count equals 4; n equals out of scope. Watch window shows Count=4.

I DO — WORKED EXAMPLE

Follow the worked method

A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.

Predict first

Step Into Add and inspect the local points and the field Points.

Model the reasoning

The local becomes 8 while the object field remains 3; the assignment targets the wrong variable.

Model the reasoning

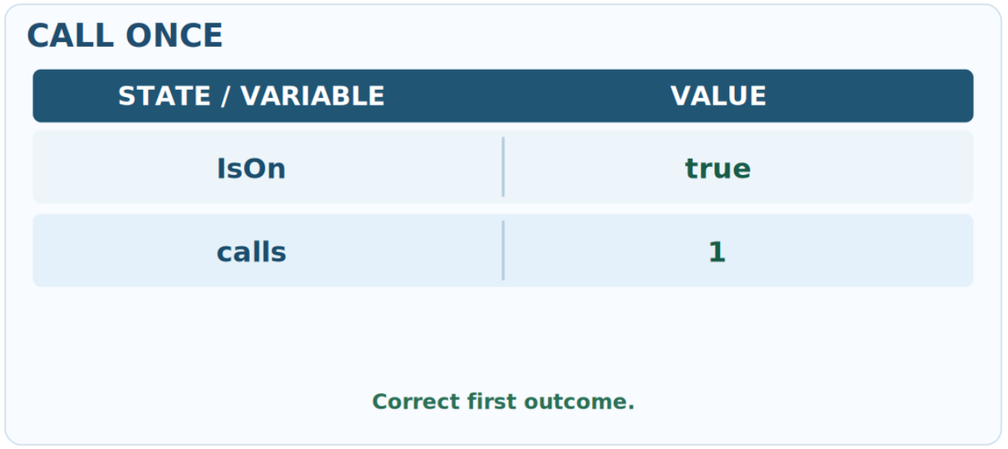
Replace it with Points += amount; and rerun. The field should be 8.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Lamp starts IsOn=false. Toggle() sets IsOn=true on every call. Diagnose the second-call error.

After the first call, true is correct; after the second, IsOn should become false.



CALL ONCE IsOn equals true; calls equals 1. Correct first outcome.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Step Over a method call means it is not executed.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.

Step Into Add and inspect the local points and the field Points.

BEFORE

STATE / VARIABLE	VALUE
Points	3
amount	5

Expected final Points=8.

BEFORE Points equals 3; amount equals 5. Expected final Points=8.

A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.

Step Into Add and inspect the local points and the field Points.

INSIDE Add

STATE / VARIABLE	VALUE
points local	8
Points field	3

Watch field does not change.

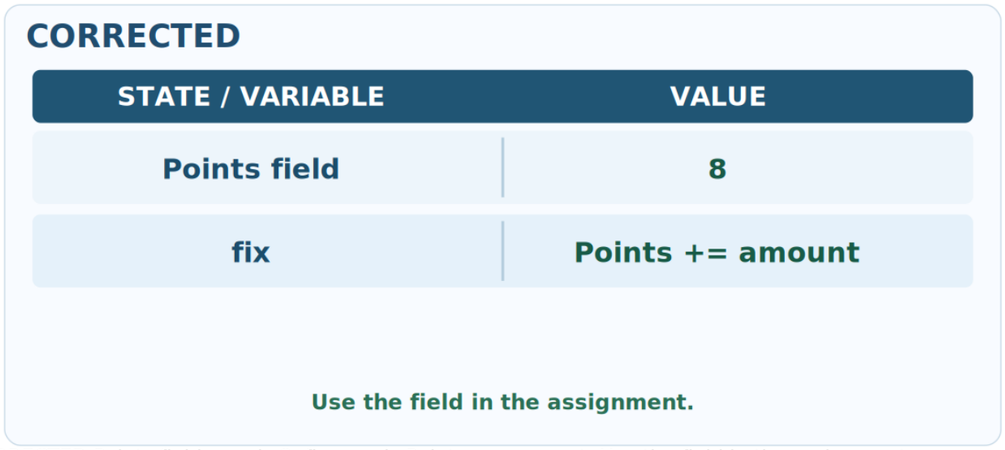
INSIDE Add points local equals 8; Points field equals 3. Watch field does not change.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.

Step Into Add and inspect the local points and the field Points.



CORRECTED Points field equals 8; fix equals Points += amount. Use the field in the assignment.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design an IDE-debugging workflow for a temperature sensor whose Readings list gains two entries after a single AddReading() call.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain Step Into and Step Over at a method call.

Question 2

A class method Increase(int n) contains `int Total = Total + n;` while Total is an existing property. Identify the issue and write correct C# code.

WE DO — GUIDED PRACTICE

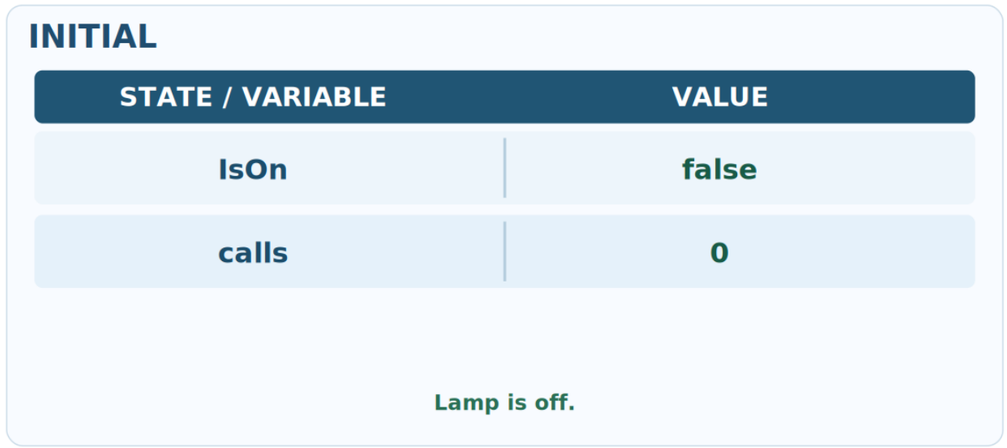
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Lamp starts IsOn=false. Toggle() sets IsOn=true on every call. Diagnose the second-call error.

After the first call, true is correct; after the second, IsOn should become false.



INITIAL IsOn equals false; calls equals 0. Lamp is off.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first After the first call, true is correct; after the second, IsOn should become false.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.